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Course: 21LEM202T - Universal Human Values - II

Unit – 4 : Session – 8 | Water, Food and Population — Right Utilisation

Part C — Long-Answer Questions (Solutions)

1. Q1. INDIA HAS WATER ABUNDANCE, MISMANAGED: India receives about 2,925 cubic km of rainfall annually — among the highest for any nation — while our total use is only around 230 cubic km; beneath us lie aquifers estimated around 433,000 cubic km, fed by forests, wetlands and rivers for millennia. The CAUSES OF MISMANAGEMENT: (1) destruction of NATURAL STORAGE — forests felled, wetlands filled, aquifers over-drawn and not recharged (59% of aquifers already declining as of 2014; about 1,615 blocks — 29% — were stressed, projected 60% critical by 2028); (2) replacement by UNNATURAL STORAGE and transport — big dams and canals — which lose water to evaporation, displacement and politics; (3) unlimited, unassessed consumption driven by the assumption 'more is better'. So scarcity is manufactured by conduct, not given by nature — the remedy is protection of natural storage, right utilisation, and human education.
2. Q2. FOOD INSECURITY IS A PROBLEM OF RIGHT UTILISATION, NOT PRODUCTION: humanity produces roughly 4.2 billion tons of food — about SIX TIMES the requirement — yet around 1.3 BILLION TONS is wasted (FAO estimate) while millions stay hungry. The animal-industry figures show the distortion: producing 1 kg of meat consumes about 14–16 kg of grain and ~21,000 litres of water; the grain fed to cattle could feed about 8.7 billion people; a meat-based diet needs ~3¼ acres per person versus ~0.2 acres for a plant-based diet; and about ONE-THIRD of all food is wasted. Clearly the Earth already produces enough; what is missing is right assessment of need, just distribution, and conduct free of greed and waste — again, a question of human understanding, not agricultural technology alone.
3. Q3. 'POPULATION WITHOUT RIGHT UNDERSTANDING': population becomes a problem not by number but by the condition of the human beings. THREE FACTORS make it so: (1) KAAM UNMAD (obsession with sensual gratification) — reproduction driven by sensation rather than responsibility, without the understanding of family as the unit of mutual fulfilment; (2) BREAKDOWN OF RELATIONSHIP — when families lack trust and justice, children are born into insecurity, and people seek compensation in consumption and further irresponsibility; (3) MEDICAL SCIENCE WITHOUT SANSKAR — technology that sustains life and reduces mortality but is used in a society that has not developed the understanding of right utilisation and responsibility, so numbers grow while conduct remains inhuman. With right understanding, the same population lives prosperously within nature's limits; without it, any number is 'too many'.
4. Q4. NATURAL STORAGE — forests, aquifers, rivers and glaciers — stores water in harmony with the whole system: forests recharge aquifers slowly, rivers distribute seasonally, glaciers release gradually; they sustain biodiversity while storing. UNNATURAL STORAGE — big dams and canals — concentrates water against the natural gradient of the land: massive evaporation, siltation, displacement of people and ecosystems, and political conflict over sharing. The lesson is not 'never build' but that natural storage, protected and enriched, is the primary infrastructure; human-made storage must supplement it within the limits of right utilisation.